# 1st Normal Form

A relation is in first normal form if and only if all underlying domain contain atomic values only. Each cell should contain only one value and relation does not contain any repeating groups. A repeating group is an attribute in which data items may repeat in each tuple at variable number of times. And also All entities must have a key, composed of an attribute or combination of attributes which uniquely identify one occurence of the entity.

**Example:** For example we have Student Relation.

| Primary → Roll No | Name | Course |
|---|---|---|
| 01 | Rashid | C \| C++ |
| 02 | Shahid | Java |
| 03 | Zahid | C \| DBMS |

it is not in first normal form because Course attributes have reapting groups.

**Method 1st:** in the first step we normalize this as by removing the repeating groups as

use for composite key example

Student  Composite Key

| Roll No | Name | Course |
|---|---|---|
| 01 | Rashid | C |
| 01 | Rashid | C++ |
| 02 | Shahid | Java |
| 03 | Zahid | C |
| 03 | Zahid | DBMS |

So here in first step we Normalize Relation in 1-NF here there is no repeating group and also Unique key like Composit key ~~Ent~~ identify Student relation.

2nd step : Method

Nar

| Roll No | Name | Coursel | Course 2 |
|---|---|---|---|
| 01 | Rahid | C | C++ |
| 02 | Shahid | Java | Null |
| 03 | Zahid | C | DBMS |

here ~~allow~~ also two condition are meet of 1-NF 1st : Roll NO Uniquely idetify the student relation. No repeating group occure. Here we see relation is M:M because A student has many Course and Course may be studied to many student.

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## 3rd ~~Normal~~ Method:

in this we make two relation base table and child. Because when M:N relation accure we also make two ~~tab~~ Entity

Student

| Roll No | Name |
|---|---|
| 01 | Rashid |
| 02 | Shahid |
| 03 | Zahid |

→ Primary

base

Course

| RollNo | Course |
|---|---|
| 01 | C |
| 01 | C++ |
| 02 | Java |
| 03 | C |
| 03 | DBMS |

→ forign key

child.